Kepir *et al. BMC Sports Science, Medicine and Rehabilitation* (2025) 17:319
https://doi.org/10.1186/s13102-025-01336-7

BMC Sports Science, Medicine and Rehabilitation

RESEARCH Open Access

Changes in eccentric hamstring strength and muscle soreness in younger and older male football players following match play

Ersagun Kepir[1], Julian Andrew Owen[1*], Gokhan Mehmet Karatay[2], Bekir Car[3], Esedullah Akaras[4], Necmiye Un Yildirim[5] and Gokhan Yagiz[6,7*]

Abstract

Introduction Hamstring strains commonly occur late in football matches. Hamstring injury risk rises with age and fatigue. This study investigated eccentric hamstring strength and perceived hamstring soreness after a football match in younger and older players.

Methods Maximal eccentric hamstring strength and perceived soreness were assessed in younger ($n = 11$, 20.2 ± 1.1 years, total playing time: 34.5 ± 8.8 min) and older ($n = 10$, 27.4 ± 3.7 years, total playing time: 44 ± 10.5 min) professional male football players before and after a friendly match aimed at selecting the primary team for the upcoming season. A two-way mixed model ANOVA was employed for statistical analyses. Furthermore, Hedges' (adjusted) *g* effect sizes were calculated to evaluate the magnitude of effects.

Results The older group experienced significant reductions in absolute (-20 *N* ± 38.5, $g = 0.58$, $p = 0.027$) and relative (-0.26 N/kg ± 0.5, $g = 0.43$, $p = 0.038$) maximal eccentric hamstring strength, while younger group showed no changes (absolute: 3 *N* ± 36.1, $g = 0.07$, $p = 0.773$; relative: -0.037 N/kg ± 0.54, $g = 0.07$, $p = 0.783$). There was no statistically significant difference between groups for the absolute ($p = 0.195$; $g = 0.45$) and relative ($p = 0.226$, $g = 0.43$) eccentric strength changes. On the other hand, both groups reported increases in perceived hamstring soreness after the match (older: 29% ± 14.1, $p < 0.001$; younger: 40% ± 14.9, $p < 0.001$), with the younger group reporting more significantly higher soreness changes than the older group (11% ± 17.2, $p = 0.025$). A significant negative association existed between age and eccentric strength changes ($R = -0.447$, $R^2 = 0.2$, $p = 0.007$).

Conclusions The older group showed significant declines in eccentric hamstring strength after the match, with age as a predictor. Specialists should focus on enhancing eccentric strength in older players and monitor their recovery to minimise injury risk. Future psycho-physiological research should examine muscle soreness perceptions and force production in fatigued younger and older football players. The results must be interpreted with caution due to the shorter playing time, unaccounted match workload, and limited sample size. Moreover, future research is necessary,

*Correspondence:
Julian Andrew Owen
j.owen@bangor.ac.uk
Gokhan Yagiz
gokhan.yagiz@amasya.edu.tr; drgokhanyagiz@gmail.com

Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

including workload and performance metrics over a longer play duration for each player, with a larger sample size to achieve more reliable conclusions.

Keywords Hamstring tears, Posterior thigh, Lower limb, Soccer, Strain injuries, Injury prevention

Introduction

Hamstring strain injuries (HSI) are the most common non-contact injuries in football (24%) [1]. There is also a high recurrence rate (16%), with recurrent injuries being more severe, resulting in a greater loss of time (3.4–3.6 days, $p = 0.003–0.01$) for training and competition, respectively [2]. Despite the significant efforts in strategies to reduce the rates of hamstring strain injuries [3–7], the incidence and burden of HSI in football have continued to rise over the past two decades [1].

The incidence of HSIs is greater during match play than during training (3.7 vs. 0.43 HSI per 1,000 h for match play and training, respectively) [8]. During match play, HSI most commonly occurs at the end of each half of the match [9–11]. This implies that muscular fatigue could be a contributing factor to HSI in football [12–14]. In light of this information, research has concentrated on monitoring the risk of HSIs during match play [15–36].

Sports injuries include complexity due to interactions between risk factors, and sports injury prevention strategies pay attention to identifying risk profiles (from risk factors to risk patterns) and evaluating multidirectional and interconnected interactions between risk factors of injuries involving the sports injuries' complex nature [37]. Consequently, research has concentrated on examining the interrelationships between injury risks to discern risk patterns [37]. Regarding the HIS, the risk factors include, but are not restricted to, fatigue [4, 38–41], lower eccentric strength qualities [42–45], and older age (having completed 23 years old or being older in football) [46, 47].

Woods and colleagues previously noted that football players aged 22 and younger exhibited a lower risk of sustaining HSI [46]. On the other hand, lower eccentric strength of the hamstrings is recognised as a risk factor for HSI, due to the insufficient eccentric action to control the tibia and antagonist muscles during the late-swing phase of running, which is identified as the most vulnerable period for the hamstrings [47]. Similarly, fatigue, which may potentially be caused by poor endurance in generating muscle force over time, is considered another risk factor for the HSI [4, 38–41]. In light of the information above, studies focused on alterations in risk factors due to fatigue, which is another risk factor of HSI [15–25, 27–29, 32–34, 36, 48].

It was stated that older amateur football players demonstrate lower eccentric hamstring strength [49]. However, it remains unclear what the effects are and how significantly age influences changes in hamstring strength qualities because of football match-play in professional settings. This is important so that practitioners may identify the potential influence of eccentric hamstring fatigue on the risk of HSIs. To the authors' knowledge, no study has examined alterations in the eccentric maximal hamstring strength after an actual football match based on age groups.

The present study aimed to investigate the football-specific fatigue leading to changes in maximal eccentric hamstring strength after a football match in younger and older players. This research seeks to provide insights into the interactions among the given risk factors for HSI during a football match. Furthermore, the study aims to identify the extent of muscle soreness [50] immediately following a football match in both younger and older players, addressing the existing research gaps in the literature. This study hypothesised that older participants would demonstrate greater decreases in maximal eccentric hamstring strength compared to younger players and would report elevated levels of hamstring muscle soreness, noting that age was identified as a risk factor for the HSI.

Methodology

Study design and participants

A 2 × 2 repeated measures design was used to analyse how the football match affected each group and compare these effects between groups. The required sample size for this study was calculated using F tests, ANOVA: Repeated measures, within-between interaction function in the G*Power software version 3.1 [51]. Effect size (ES = 0.9) was taken from a recent study [48], which investigated the effects of a 90-minute football match on eccentric hamstring strength. Nonetheless, the present study employed a 0.3 effect size, which indicates a small effect size in professional athletes [52], to ensure a better statistical power for the required sample size. The following additional parameters were used during the sample size calculation: 0.05 alpha level, 0.80 power, two groups, two measurement points, and 0.7 correlation among measurements [48]. As a result, the required quantity of the total sample size was calculated as 16. However, this study recruited 21 participants to ensure better statistical power (younger group = 11 participants; older group = 10 participants).

Initially, twenty-two players were recruited; however, one did not meet the inclusion criteria. Thus, twenty-one participants, all male professional footballers, took part. Eleven players aged 19 to 22 formed the younger group,

while ten aged 23 to 33 made up the older group. This age distinction considers the risks associated with being over 23 in football [46]. Ethical approval was secured from the local research ethics committee at Erzurum Technical University before the study commenced (date and number: 06/09/2023, 10 – 01). Written informed consent was obtained from all participants prior to the first session according to the Declaration of Helsinki [53]. Participants were on the same football team and experienced identical training loads. The intervention occurred during the pre-season training camp, three weeks before the competitive season. The intervention match was viewed as a selection competition, motivating players to demonstrate their abilities. Data collection was conducted between 07/09/2023 and 15/09/2023.

Inclusion criteria considered (a) being a male, (b) being a registered professional football player in the Turkish Football Federation, and (c) being aged 18 to 37. Perceived muscle soreness assessments ensured no soreness after the pre-test eccentric strength measurements on the intervention day. Prior to the study, participants' age, body mass, height, football-specific training history (total years), injury history, and training frequency (hours, sessions per week) were recorded. The dominant leg was considered the kicking leg in football [54]. Measurements were initially taken on the non-dominant leg, followed by the dominant leg. Scores for both the dominant and non-dominant legs were recorded and utilised in the statistical analyses (22 legs for the younger group and 20 legs for the older group) (Table 1). The tests and interventions were conducted on a real natural grass football pitch.

Before the first session, participants were familiarised with measurement procedures by the researchers. They engaged in a 15-minute warm-up on the pitch. Then, maximal eccentric hamstring strength was measured to establish a baseline. This was done a week before the match to counteract fatigue from previous assessments. After one week, participants played a friendly match, again performing a similar 15-minute warm-up. Hamstring muscle soreness was assessed before the match using a 1–100 scale [50]. Afterwards, participants took part in a ninety-minute football match. As it was pre-season with no limits on substitutions, players were rotated to assess performance. Participants' total playing time was recorded using a chronometer for each participant and averaged for each group (Table 1). After the match, participants underwent post-test measurements for perceived hamstring soreness and maximal eccentric hamstring strength, completed in under three minutes after play, to identify any decrements in strength.

Warm-up

The warm-up programme comprised twelve football-specific activities carried out on a 20-metre shuttle on the pitch, which included: light jogging, side-stepping, backwards jogging, forward and backwards skipping with arm circles, jumping jacks, high kicks, high knees, dynamic hamstring stretches, walking lunges, sprints, and high knees at increased speeds [48].

Measurement procedures

All pre-test and post-test measurements were conducted first on the non-dominant leg and then on the dominant leg. Scores for both the dominant and non-dominant legs were recorded. A handheld strength dynamometer was used to assess the maximal eccentric hamstring strength, following the instructions of Goosens et al. [44]. This measurement previously indicated high to very high reproducibility [44, 48]. However, due to the user dependence of the measurement, an intra-rater intra-day reproducibility assessment was carried out for this study.

The participant laid down on a medical bed in a prone position. The hip was positioned neutrally (0 degrees of extension or flexion). Participants kept their hands behind their backs to minimise compensation. The knee angle was at 60 degrees of flexion in the starting position. A goniometer was employed to confirm the positions of the hip and knee. Subsequently, the tester exerted maximal force in the direction of full knee extension using a handheld dynamometer (Lafayette Manual Muscle Test System™ (Model 01165) (Lafayette Instrument Company, Lafayette, Indiana)) [44]. Before the measurement, participants were informed that the assessor would extend their leg fully and that they should resist as much as possible during the measurement. There was a 30-second rest period between the two repetitions. Each test was

Table 1 Participants' characteristics

	Age (years) (mean ± standard deviation)	Stature (cm) (mean ± standard deviation)	Body Mass (kg) (mean ± standard deviation)	Total years of playing football (years) (mean ± standard deviation)	Total Playing Time on the football Match on the Intervention Day (minutes) (mean ± standard deviation)
Younger group ($n=11$)	20.2 ± 1.1	180 ± 6.1	71.1 ± 5.8	11 ± 3.5	34.5 ± 8.8 (Between 30 and 45 min)
Older group ($n=10$)	27.4 ± 3.7	182 ± 4.6	76.7 ± 4.6	15.5 ± 3.1	44 ± 10.5 (Between 30 and 60 min)
p-value for the difference	**Age: p** < 0.001**	Stature: $p=0.399$	Body mass: $p=0.84$	Total years of playing: $p=0.823$	Total match playing time: $p=0.934$

completed in approximately five seconds to ensure consistent velocity between the measurements. Each leg was measured twice by the same tester (the first author), and the highest score was accepted as the maximal eccentric strength of the hamstring [44, 48].

Perceived muscle soreness was assessed before and after the match for both legs via an analogue scale ranging from 0 to 100 points [50], which refers to 100, "the worst pain"; 75, "severe pain"; 50, "moderate pain"; 25, "mild pain"; 0, "no soreness" [50]. Participants were asked to report their soreness score for each hamstring muscle group of the dominant and the non-dominant legs.

Statistical analyses

The primary statistical analyses of the results were conducted using the IBM SPSS 29.0 software (IBM Corp, Armonk, NY). Participants' characteristics (age, stature, and body mass) were presented as means and standard deviations and assessed using a one-way ANOVA. A two-way mixed model ANOVA with Bonferroni correction was employed to analyse the following dependent variables: hamstring muscle soreness scores (two time points for each group) and eccentric maximal hamstring strength (two time points for each group). The ICC values for maximal eccentric hamstring strength were calculated for two-way random, absolute agreements based on average measurements for the reproducibility analysis of both legs, as well as the average score [55]. ICC values indicate the following: an ICC value of less than 0.5 signifies poor reproducibility, an ICC value between 0.5 and 0.75 indicates moderate reproducibility, an ICC value ranging from 0.75 to 0.9 reflects high reproducibility, and an ICC value exceeding 0.9 denotes very high reproducibility [56]. Additionally, the Hedges' (adjusted) *g* effect sizes were calculated. The intervention's effect size has been graded as follows: small (0.2), medium (0.5), or large (0.8), which are commonly employed for Cohen's *d* [57] and Hedges' *g* [58] effect size classifications [59].

Results

Participants' characteristics

Twenty-one players (11 younger (20.2 ± 1.1 years old) and 10 older (27.4 ± 3.7 years old)) male football players completed the study protocols. The characteristics of the participants are outlined in Table 1. The intervention in this study comprised a 90-minute competitive friendly match. Due to the selection process for the primary team, the coach aims to include and observe as many players as possible, which led to lesser playing times for both groups (younger group: 34.5 ± 8.8 min; older group: 44 ± 10.5 min) (Table 1). Out of the twenty participants, nineteen were right-leg dominant, while one player from the younger group was left-leg dominant. There were two goalkeepers: one from the younger group and the other from the older group.

Reproducibility of the maximal eccentric hamstring strength measurements

Due to missing data, 31 leg measurements were included in the reproducibility analyses. An imputation method was not employed to address any missing data, and they were directly excluded from the analyses. High reproducibility was identified for the maximal eccentric hamstring strength measurements using a handheld dynamometer ($n = 31$, ICC = 0.888 [0.771, 0.946]).

Maximal eccentric hamstring strength and perceived muscle soreness

While no significant change in eccentric hamstring strength was observed in the younger group ($p = 0.773$, g = 0.07), the older group experienced a significant post-match reduction ($p = 0.027$, g = 0.58) (Table 2). There was no statistically significant time* group interaction for the eccentric strength changes ($p = 0.195$; $g = 0.45$). When maximal eccentric hamstring strength relative to body mass was examined, the younger group did not show a significant decrease ($p = 0.783$, $g = 0.07$). The older

Table 2 Absolute maximal eccentric hamstring strength (Newton) and perceived muscle soreness results (Percentage)

	Pre-test (M ± SD)	Post-test (M ± SD)	Post-test and pre-test difference (M ± SD)	The *p*-value for the post-test and pre-test difference	Pre-test for the Perceived Muscle Soreness (M ± SD)	Post-test for the Perceived Muscle Soreness (M ± SD)	Post-test and pre-test difference (M ± SD)	The *p*-value for the post-test and pre-test difference
Younger group	276.7 ± 41.6 ($n = 15$)	273.9 ± 40.8 ($n = 15$)	-3 ± 36.1 ($n = 15$), Hedges' $g = 0.07$	$p = 0.773$	21.7 ± 3.9 ($n = 18$)	61.7 ± 4.3 ($n = 18$)	40 ± 14.9 ($n = 18$), Hedges' $g = 9.74$	**p** < 0.001**
Older group	276.9 ± 9.4 ($n = 19$)	257 ± 48 ($n = 19$)	-20 ± 38.5 ($n = 19$) Hedges' $g = 0.58$	**p* = 0.027**	17 ± 3.6 ($n = 20$)	46 ± 4 ($n = 20$)	29 ± 14.1 ($n = 20$), Hedges' $g = 7.62$	**p** < 0.001**
p-value for the time* group interactions and ES	$p = 0.195$; Hedges' g = 0.45				**p* = 0.025**, Hedges' $g = 0.76$			

Abbreviations: ES, Effect size for changes from baseline between groups; M, mean; SD, standard deviation* significantly different; ** significantly different

Table 3 Relative maximal eccentric hamstring strength to body mass (Newton/kg)

	Pre-test (M ± SD)	Post-test (M ± SD)	Post-test and pre-test difference (M ± SD)	The *p*-value for the post-test and pre-test difference
Younger group	3.99 ± 0.61 (*n* = 15)	3.95 ± 0.59 (*n* = 15)	-0.037 ± 0.54, Hedges' *g* = 0.07 (*n* = 15)	*p* = 0.783
Older group	3.62 ± 0.55 (*n* = 19)	3.36 ± 0.66 (*n* = 19)	-0.26 ± 0.5, Hedges' *g* = 0.43 (*n* = 19)	**p* = 0.038**
p-value for the time* group interactions and ES	*p* = 0.226, Hedges' *g* = 0.43			

Abbreviations: ES, Effect size for changes from baseline between groups; M, mean; SD, standard deviation* significantly different; ** significantly different

group showed a significant decrement in relative maximal eccentric hamstring strength ($p = 0.038$, g = 0.43) (Table 3). No statistically significant time*group interaction was found regarding relative maximal eccentric hamstring strength changes ($p = 0.226$, Hedges' $g = 0.43$). On the other hand, there were no differences between the legs regarding the changes from the pre-test in maximal eccentric hamstring strength (D: -11 ± 33.3 vs. ND: -13.1 ± 42.3, $p = 0.871$) and soreness (D: 33.9 ± 15 vs. ND: 34.5 ± 16.1, $p = 0.917$).

A significant negative association was observed between age and strength change (R = -0.447, R^2 = 0.20, $p = 0.007$), suggesting that strength losses were greater in older players. Despite older players (44 ± 10.5 min) playing for a longer duration than the younger group (34.5 ± 8.8 min) during the game (Table 1), there were no significant regressions detected between the playing time and maximal eccentric strength alterations ($R = 0.238$, $R^2 = 0.057$, $p = 0.175$). There was a significant time* group interaction in the perceived muscle soreness results ($p = 0.025$). Both groups reported significantly increased soreness post-match; however, younger players exhibited a greater relative increase than their older counterparts (Table 2).

Discussion

This study found significant moderate eccentric strength decrements (ES = 0.58, -7%) in the older group after 44 ± 10.5 min of match-play. Previous studies also reported decreases in eccentric hamstring strength after a football match [15–36]. For example, Bueno et al. [15] observed significant eccentric strength decreases in hamstrings immediately after (ES = 1.1, -12%) an actual football match-play (14 players played 90 min, one player played 85 min) in professional football players with a closer age group (25.73 ± 4.14 years old) to our older group (27.4 ± 3.7 years old), in light of the information that significant reductions in eccentric hamstring strength following a full-time football match, as pointed out by Bueno et al. [15], our findings suggest that medium decrements in eccentric hamstring strength of our older group after approximately a half-time football match-play can be expected. A future study should investigate whether there are potentially greater reductions in hamstring eccentric strength in both groups following ninety minutes of a football match. Practitioners should consider enhancing the maximal eccentric strength of older football players and developing optimal recovery strategies that include hamstrings.

Ageing and/or previous injuries may deteriorate the structure and functions of human skeletal muscles [7, 60–69]. It is well established in the literature that ageing adversely affects synaptic input production, muscle motor-unit discharge rates and force production capacity, which can trigger muscle fibre denervation that increases the fibres not contributing to the force production [64–67, 70, 71]. Older adults show less frequent motor neuron firing than younger adults, and this difference even increases at higher intensity muscle contraction [67]. Therefore, the findings of this study could be reasonable in that the older group showed significant decrements in eccentric hamstring strength, while the younger group did not show any. This result could suggest that older football players should give importance to training to improve the sustainability of the eccentric strength of the hamstrings during a football match. Older players may consider adapting eccentric training modalities to maintain the sustainability of the eccentric strength and eccentric strength improvements in hamstrings using the Askling L protocol [72], Nordic hamstring [3, 5, 6], Russian Belt exercise [73], and single-leg hamstring bridge [43] exercises. Future studies should focus on the effects of eccentric hamstring training on the endurance of the hamstrings and the neural and motor physiology of the hamstring muscle during a football match.

An interesting finding from the study was that the younger group reported higher perceived muscle soreness following half of a football match compared to the older group. Despite this increase in soreness, this did not correspond to a similar pattern of decrease in the hamstring eccentric strength. Indeed, both younger and older groups did not play the entire 90 min or beyond the time (extra-times) in the study due to the nature of a friendly football match, which is a limitation of this study,

as it does not see the full effects of a football match on the hamstrings, and this should be investigated by a future study. Interestingly, the perception of muscle soreness may be different between the older and younger groups, or the younger group may be able to produce higher force at fatigue compared to the older group. Future psychophysiological research should reveal the reason behind these findings.

This study has several limitations. Firstly, the football players did not play for ninety minutes due to player substitutions, which were a preseason requirement at a friendly match aimed at monitoring multiple players by the coaches. Secondly, while the measurement of hamstring maximal eccentric strength demonstrates high reproducibility, it remains a user-dependent technique that can introduce unaccounted variability to the outcome measures. Thirdly, the presence of 21 participants may be regarded as a small sample size, thereby limiting the ability to draw general conclusions relevant to the broader football community. This should be recognised as a limitation that highlights the need for future studies involving a larger sample size.

Additionally, this study did not record information about participants' running distances, and this is another confounding factor of the present study. Moreover, some players did not attend to some of the baseline or post-test measurements, and their data could not be used in statistical analyses, which prevented this study from reaching higher statistical power. Furthermore, the previous training history and match external outputs are unknown, and previous injury history and playing positions are not homogeneous between the groups, which are other limitations of this study. The findings of this study were based on a single football club and a single game, which is a limitation of the generalisability of this study. Future studies should encompass multiple teams, control for player workload, and utilise larger samples to enhance generalisability. Even though the differences in match duration were not statistically significant between the groups, the total playing time of both groups was shorter than that commonly reported in the literature. This may limit comparability between the findings of this study and other studies.

This study was conducted during the first two weeks of pre-season training, which may affect outcome measures due to potential deconditioning since the previous season's end. Additionally, players' muscle soreness was assessed during a single game. Future research should explore muscle soreness perception longitudinally for deeper insights. The measurement techniques used may face challenges in player adherence to pre- and post-test protocols in a longitudinal design, which could limit future studies investigating these outcome measures. This study lacked half-time assessments, a common occurrence for hamstring strain injuries (HSI) at the end of halves, warranting evaluation in future research. Furthermore, it's unclear if social factors influence players' perceived muscle soreness, presenting another limitation of this study. The absence of performance or workload metrics (e.g., GPS data) is another limitation, making it challenging to draw stronger conclusions for this study. Despite the inclusion of one goalkeeper in the statistical analyses for each group, the involvement of goalkeepers in these analyses may introduce another confounding factor due to their differing match demands compared to outfield players.

Conclusions

The older group showed moderately significant reductions in maximal eccentric hamstring strength after half a football match. Sports specialists should focus on improving the reduced eccentric strength of the hamstrings in older players, paying particular attention to monitoring their recovery to baseline levels, as this may increase the risk of injury in subsequent matches. However, the results should be interpreted with caution due to the short playing time, unmeasured match workload, and small sample size. Furthermore, future studies that include workload and performance metrics over a longer duration of play per player, involving a larger sample size, are crucial for drawing more robust conclusions.

Abbreviations

ACL	Anterior cruciate ligament
ANOVA	Analysis of variance
HSIs	Hamstring strain injuries
ICC	Interclass correlation coefficient
SLHB	Single-leg hamstring bridge

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13102-025-01336-7.

Supplementary Material 1

Acknowledgements

The Ministry of National Education of the Turkey funds the first author, Kepir, for his postgraduate studies. A version of this study will also be published as a chapter of Kepir's PhD thesis.

Author contributions

E.K.: Research design, recruitment of the participants, preparing the data collection settings, data collection, data analysis, and writing the first draft of the manuscript. J.A.O.: Research design, editing the manuscript, and supervision. G.M.K: Recruitment of the participants, preparing the data collection settings, and data collection. B.C.: Recruitment of the participants, preparing the data collection settings, and data collection. E.A.: Research design, preparing the data collection settings, and data collection. N.U.Y: Recruitment of the participants, preparing the data collection settings, and data collection. G.Y.: Research design, writing the first draft of the manuscript, editing the manuscript, data analysis and supervision.

Funding

This study did not receive any specific funding.

Data availability
All the data are presented in the manuscript and Supporting Information 1. Data.pdf.

Declarations

Ethics approval and consent to participate
Informed consent was obtained from participants before the first session, complying with the Declaration of Helsinki. Before commencing the study, ethical approval was secured from the local research ethics committee at Erzurum Technical University (date and number: 06/09/2023, 10 – 01).

Consent for publication
All the participants provided written consent for publication.

Competing interests
The authors declare no competing interests.

Clinical trial number
Not applicable.

Author details
[1]Institute for Applied Human Physiology, School of Psychology and Sport Science, Bangor University, Bangor, UK
[2]Department of Physiotherapy and Rehabilitation, Faculty of Health Sciences, Gazi University, Ankara, Turkey
[3]Faculty of Sport Sciences, Bandirma Onyedi Eylul University, Balikesir, Turkey
[4]Department of Physiotherapy and Rehabilitation, Faculty of Health Sciences, Erzurum Technical University, Erzurum, Turkey
[5]Faculty of Physiotherapy and Rehabilitation, Health Sciences University, Ankara, Turkey
[6]Department of Physiotherapy and Rehabilitation, Faculty of Health Sciences, Amasya University, Amasya, Turkey
[7]Department of Physical Therapy and Rehabilitation, Faculty of Health Sciences, Tokyo Metropolitan University, Tokyo, Japan

Received: 8 April 2025 / Accepted: 28 August 2025
Published online: 05 November 2025

References

1. Ekstrand J, Bengtsson H, Waldén M, Davison M, Khan KM, Hägglund M. Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: the UEFA elite club injury study from 2001/02 to 2021/22. Br J Sports Med. 2022;bjsports–2021–105407. https://doi.org/10.1136/bjsports-2021-105407
2. Ekstrand J, Krutsch W, Spreco A, van Zoest W, Roberts C, Meyer T, et al. Time before return to play for the most common injuries in professional football: a 16-year follow-up of the UEFA elite club injury study. Br J Sports Med. 2020;54(7):421–6. https://doi.org/10.1136/bjsports-2019-100666
3. Arnason A, Andersen TE, Holme I, Engebretsen L, Bahr R. Prevention of hamstring strains in elite soccer: an intervention study. Scand J Med Sci Sports. 2008;18(1):40–8. https://doi.org/10.1111/j.1600-0838.2006.00634.x
4. Opar DA, Williams MD, Shield AJ. Hamstring strain injuries: factors that lead to injury and re-injury. Sports Med. 2012;42(3):209–26. https://doi.org/10.2165/11594800-000000000-00000
5. Petersen J, Thorborg K, Nielsen MB, Budtz-Jørgensen E, Hölmich P. Preventive effect of eccentric training on acute hamstring injuries in men's soccer: a cluster-randomized controlled trial. Am J Sports Med. 2011;39(11):2296–303. https://doi.org/10.1177/0363546511419277
6. van der Horst N, Smits DW, Petersen J, Goedhart EA, Backx FJ. The preventive effect of the nordic hamstring exercise on hamstring injuries in amateur soccer players: a randomized controlled trial. Am J Sports Med. 2015;43(6):1316–23. https://doi.org/10.1177/0363546515574057
7. Yagiz G, Akaras E, Kubis H-P, Owen JA. Heterogeneous effects of eccentric training and nordic hamstring exercise on the biceps femoris fascicle length based on ultrasound assessment and extrapolation methods: a systematic review of randomised controlled trials with meta-analyses. PLoS ONE. 2021;16(11):e0259821. https://doi.org/10.1371/journal.pone.0259821
8. Ekstrand J, Hägglund M, Waldén M. Epidemiology of muscle injuries in professional football (Soccer). Am J Sports Med. 2011;39(6):1226–32. https://doi.org/10.1177/0363546510395879
9. Ekstrand J, Hägglund M, Waldén M. Injury incidence and injury patterns in professional football: the UEFA injury study. Br J Sports Med. 2011;45(7):553–8. https://doi.org/10.1136/bjsm.2009.060582
10. Hawkins RD, Hulse MA, Wilkinson C, Hodson A, Gibson M. The association football medical research programme: an audit of injuries in professional football. Br J Sports Med. 2001;35(1):43–7. https://doi.org/10.1136/bjsm.35.1.43
11. Woods C, Hawkins RD, Maltby S, Hulse M, Thomas A, Hodson A. The football association medical research programme: an audit of injuries in professional football–analysis of hamstring injuries. Br J Sports Med. 2004;38(1):36–41. https://doi.org/10.1136/bjsm.2002.002352
12. McCall A, Carling C, Nedelec M, Davison M, Le Gall F, Berthoin S, et al. Risk factors, testing and preventative strategies for non-contact injuries in professional football: current perceptions and practices of 44 teams from various premier leagues. Br J Sports Med. 2014;48(18):1352–7. https://doi.org/10.1136/bjsports-2014-093439
13. Meurer MC, Silva MF, Baroni BM. Strategies for injury prevention in Brazilian football: perceptions of physiotherapists and practices of premier league teams. Phys Ther Sport. 2017;28:1–8. https://doi.org/10.1016/j.ptsp.2017.07.004
14. Moreno-Perez V, Paredes V, Pastor D, Garrosa FN, Vielcazat SJ, Del Coso J, et al. Under-exposure to official matches is associated with muscle injury incidence in professional footballers. Biol Sport. 2021;38(4):563–71. https://doi.org/10.5114/biolsport.2021.100360
15. Bueno CA, de Araujo Ribeiro-Alvares JB, Oliveira GS, Grazioli R, Veeck F, Pinto RS, et al. Post-match recovery of eccentric knee flexor strength in male professional football players. Phys Ther Sport. 2021;47:140–6. https://doi.org/10.1016/j.ptsp.2020.11.032
16. Cohen DD, Zhao B, Okwera B, Matthews MJ, Delextrat A. Angle-specific eccentric hamstring fatigue after simulated soccer. Int J Sports Physiol Perform. 2015;10(3):325–31. https://doi.org/10.1123/ijspp.2014-0088
17. Coratella G, Bellin G, Beato M, Schena F. Fatigue affects peak joint torque angle in hamstrings but not in quadriceps. J Sports Sci. 2015;33(12):1276–82. https://doi.org/10.1080/02640414.2014.986185
18. de Abreu Camarda SR, Denadai BS. Does muscle imbalance affect fatigue after soccer specific intermittent protocol? J Sci Med Sport. 2012;15(4):355–60. https://doi.org/10.1016/j.jsams.2011.11.257
19. Delextrat A, Baker J, Cohen DD, Clarke ND. Effect of a simulated soccer match on the functional hamstrings-to-quadriceps ratio in amateur female players. Scand J Med Sci Sports. 2013;23(4):478–86. https://doi.org/10.1111/j.1600-0838.2011.01415.x
20. Delextrat A, Gregory J, Cohen D. The use of the functional H:Q ratio to assess fatigue in soccer. Int J Sports Med. 2010;31(3):192–7. https://doi.org/10.1055/s-0029-1243642
21. Delextrat A, Piquet J, Matthews MJ, Cohen DD. Strength-endurance training reduces the hamstrings strength decline following simulated football competition in female players. Front Physiol. 2018;9:1059. https://doi.org/10.3389/fphys.2018.01059
22. Grazioli R, Lopez P, Andersen LL, Machado CLF, Pinto MD, Cadore EL, et al. Hamstring rate of torque development is more affected than maximal voluntary contraction after a professional soccer match. Eur J Sport Sci. 2019;19(10):1336–41. https://doi.org/10.1080/17461391.2019.1620863
23. Greco CC, da Silva WL, Camarda SR, Denadai BS. Fatigue and rapid hamstring/quadriceps force capacity in professional soccer players. Clin Physiol Funct Imaging. 2013;33(1):18–23. https://doi.org/10.1111/j.1475-097X.2012.01160.x
24. Greig M, Siegler JC. Soccer-specific fatigue and eccentric hamstrings muscle strength. J Athl Train. 2009;44(2):180–4. https://doi.org/10.4085/1062-6050-44.2.180
25. Jones RI, Ryan B, Todd AI. Muscle fatigue induced by a soccer match-play simulation in amateur black South African players. J Sports Sci. 2015;33(12):1305–11. https://doi.org/10.1080/02640414.2015.1022572
26. Kakavas G, Malliaropoulos N, Kaliakmanis A, Georgios B, Maffulli N. A ninety-minute football match increases hamstring flexibility in professional players. J Biol Regul Homeost Agents. 2020;34(5 Suppl 1):87–92. IORS Special Issue on Orthopedics.
27. Lehnert M, De Ste Croix M, Zaatar A, Hughes J, Varekova R, Lastovicka O. Muscular and neuromuscular control following soccer-specific exercise in male youth: changes in injury risk mechanisms. Scand J Med Sci Sports. 2017;27(9):975–82. https://doi.org/10.1111/sms.12705

28. Lehnert M, De Ste Croix M, Zaatar A, Lipinska P, Stastny P. Effect of a simulated match on lower limb neuromuscular performance in youth footballers-a two year longitudinal study. Int J Environ Res Public Health. 2020;17(22). https://doi.org/10.3390/ijerph17228579
29. Marshall PW, Lovell R, Jeppesen GK, Andersen K, Siegler JC. Hamstring muscle fatigue and central motor output during a simulated soccer match. PLoS ONE. 2014;9(7):e102753. https://doi.org/10.1371/journal.pone.0102753
30. Marshall PW, Lovell R, Siegler JC. Changes in passive tension of the hamstring muscles during a simulated soccer match. Int J Sports Physiol Perform. 2016;11(5):594–601. https://doi.org/10.1123/ijspp.2015-0009
31. Rahnama N, Lees A, Reilly T. Electromyography of selected lower-limb muscles fatigued by exercise at the intensity of soccer match-play. J Electromyogr Kinesiol. 2006;16(3):257–63. https://doi.org/10.1016/j.jelekin.2005.07.011
32. Rahnama N, Reilly T, Lees A, Graham-Smith P. Muscle fatigue induced by exercise simulating the work rate of competitive soccer. J Sports Sci. 2003;21(11):933–42. https://doi.org/10.1080/0264041031000140428
33. Small K, McNaughton L, Greig M, Lovell R. Effect of timing of eccentric hamstring strengthening exercises during soccer training: implications for muscle fatigability. J Strength Cond Res. 2009;23(4):1077–83. https://doi.org/10.1519/JSC.0b013e318194df5c
34. Small K, McNaughton L, Greig M, Lovell R. The effects of multidirectional soccer-specific fatigue on markers of hamstring injury risk. J Sci Med Sport. 2010;13(1):120–5. https://doi.org/10.1016/j.jsams.2008.08.005
35. Small K, McNaughton LR, Greig M, Lohkamp M, Lovell R. Soccer fatigue, sprinting and hamstring injury risk. Int J Sports Med. 2009;30(8):573–8. https://doi.org/10.1055/s-0029-1202822
36. Wollin M, Thorborg K, Pizzari T. The acute effect of match play on hamstring strength and lower limb flexibility in elite youth football players. Scand J Med Sci Sports. 2017;27(3):282–8. https://doi.org/10.1111/sms.12655
37. Bittencourt NFN, Meeuwisse WH, Mendonça LD, Nettel-Aguirre A, Ocarino JM, Fonseca ST. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition-narrative review and new concept. Br J Sports Med. 2016;50(21):1309–14. https://doi.org/10.1136/bjsports-2015-095850
38. Devlin L. Recurrent posterior thigh symptoms detrimental to performance in rugby union: predisposing factors. Sports Med. 2000;29(4):273–87. https://doi.org/10.2165/00007256-200029040-00005
39. Heiser TM, Weber J, Sullivan G, Clare P, Jacobs RR. Prophylaxis and management of hamstring muscle injuries in intercollegiate football players. Am J Sports Med. 1984;12(5):368–70. https://doi.org/10.1177/036354658401200506
40. Mair SD, Seaber AV, Glisson RR, Garrett WE Jr. The role of fatigue in susceptibility to acute muscle strain injury. Am J Sports Med. 1996;24(2):137–43. https://doi.org/10.1177/036354659602400203
41. Worrell TW, Perrin DH. Hamstring muscle injury: the influence of strength, flexibility, warm-up, and fatigue. J Orthop Sports Phys Ther. 1992;16(1):12–8. https://doi.org/10.2519/jospt.1992.16.1.12
42. De Vos R-J, Reurink G, Goudswaard G-J, Moen MH, Weir A, Tol JL. Clinical findings just after return to play predict hamstring re-injury, but baseline MRI findings do not. Br J Sports Med. 2014;48(18):1377–84.
43. Freckleton G, Cook J, Pizzari T. The predictive validity of a single leg bridge test for hamstring injuries in Australian rules football players. Br J Sports Med. 2014;48(8):713–7.
44. Goossens L, Witvrouw E, Vanden Bossche L, De Clercq D. Lower eccentric hamstring strength and single leg hop for distance predict hamstring injury in PETE students. Eur J Sport Sci. 2015;15(5):436–42.
45. Schuermans J, Van Tiggelen D, Danneels L, Witvrouw E. Susceptibility to hamstring injuries in soccer: a prospective study using muscle functional magnetic resonance imaging. Am J Sports Med. 2016;44(5):1276–85.
46. Woods C, Hawkins RD, Maltby S, Hulse M, Thomas A, Hodson A. The football association medical research programme: an audit of injuries in professional football—analysis of hamstring injuries. Br J Sports Med. 2004;38(1):36. https://doi.org/10.1136/bjsm.2002.002352
47. Green B, Bourne MN, van Dyk N, Pizzari T. Recalibrating the risk of hamstring strain injury (HSI): a 2020 systematic review and meta-analysis of risk factors for index and recurrent hamstring strain injury in sport. Br J Sports Med. 2020;54(18):1081. https://doi.org/10.1136/bjsports-2019-100983
48. Yagiz G, Dayala VK, Williams K, Owen JA, Kubis H-P. Alterations in biceps femoris long head fascicle length, eccentric hamstring strength qualities and single-leg hop distance throughout the ninety minutes of TSAFT90 simulated football match. PLoS ONE. 2022;17(12):e0278222. https://doi.org/10.1371/journal.pone.0278222
49. Vicens-Bordas J, Esteve E, Fort-Vanmeerhaeghe A, Clausen MB, Bandholm T, Opar D, et al. Eccentric hamstring strength is associated with age and duration of previous season hamstring injury in male soccer players. Int J Sports Phys Ther. 2020;15(2):246–53.
50. Burnett D, Smith K, Smeltzer C, Young K, Burns S. Perceived muscle soreness in recreational female runners. Int J Exerc Sci. 2010;3(3):108–16.
51. Faul F, Erdfelder E, Buchner A, Lang A-G. Statistical power analyses using G*Power 3.1: tests for correlation and regression analyses. Behav Res Methods. 2009;41(4):1149–60. https://doi.org/10.3758/BRM.41.4.1149
52. Rhea MR. Determining the magnitude of treatment effects in strength training research through the use of the effect size. J Strength Cond Res. 2004;18(4):918–20. https://doi.org/10.1519/14403.1
53. Association WM. World medical association declaration of helsinki: ethical principles for medical research involving human subjects. JAMA. 2013;310(20):2191–4. https://doi.org/10.1001/jama.2013.281053
54. Zago M, Motta AF, Mapelli A, Annoni I, Galvani C, Sforza C. Effect of leg dominance on the center-of-mass kinematics during an inside-of-the-foot kick in amateur soccer players. J Hum Kinet. 2014;42:51–61. https://doi.org/10.2478/hukin-2014-0060
55. Weir JP. Quantifying test-retest reliability using the intraclass correlation coefficient and the SEM. J Strength Cond Res. 2005;19(1):231–40. https://doi.org/10.1519/15184.1
56. Portney LG, Watkins MP. Foundations of clinical research: applications to practice. Pearson/Prentice Hall Upper Saddle River, NJ; 2009.
57. Cohen J. Statistical power analysis for the behavioral sciences. 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates; 1988.
58. Hedges LV, Olkin I. Statistical methods for meta-analysis. Academic; 2014.
59. Brydges CR. Effect size guidelines, sample size calculations, and statistical power in gerontology. Innov Aging. 2019;3(4):igz036–igz. https://doi.org/10.1093/geroni/igz036
60. Clark RA. Hamstring injuries: risk assessment and injury prevention. Ann Acad Med Singap. 2008;37(4):341–6.
61. Cruz-Jentoft AJ, Bahat G, Bauer J, Boirie Y, Bruyère O, Cederholm T, et al. Sarcopenia: revised European consensus on definition and diagnosis. Age Ageing. 2019;48(1):16–31. https://doi.org/10.1093/ageing/afy169
62. Doherty TJ. Invited review: aging and sarcopenia. J Appl Physiol (1985). 2003;95(4):1717–27. https://doi.org/10.1152/japplphysiol.00347.2003
63. Narici MV, Maganaris CN, Reeves ND, Capodaglio P. Effect of aging on human muscle architecture. J Appl Physiol (1985). 2003;95(6):2229–34. https://doi.org/10.1152/japplphysiol.00433.2003
64. Orssatto L, Wiest MJ, Diefenthaeler F. Neural and musculotendinous mechanisms underpinning age-related force reductions. Mech Ageing Dev. 2018;175:17–23. https://doi.org/10.1016/j.mad.2018.06.005
65. Orssatto LBR, Blazevich AJ, Trajano GS. Ageing reduces persistent inward current contribution to motor neurone firing: potential mechanisms and the role of exercise. J Physiol. 2023;601(17):3705–16. https://doi.org/10.1113/JP284603
66. Orssatto LBR, Borg DN, Blazevich AJ, Sakugawa RL, Shield AJ, Trajano GS. Intrinsic motoneuron excitability is reduced in soleus and tibialis anterior of older adults. Geroscience. 2021;43(6):2719–35. https://doi.org/10.1007/s11357-021-00478-z
67. Orssatto LBR, Borg DN, Pendrith L, Blazevich AJ, Shield AJ, Trajano GS. Do motoneuron discharge rates slow with aging? A systematic review and meta-analysis. Mech Ageing Dev. 2022;203:111647. https://doi.org/10.1016/j.mad.2022.111647
68. Yagiz G, Shida N, Kuruma H, Furuta M, Morimoto K, Yamada M, et al. Rugby players exhibit stiffer biceps femoris, lower biceps femoris fascicle length to knee extensors, and knee flexors to extensors muscle volume ratios than active controls. Int J Sports Physiol Perform. 2023;18(9):1030–7. https://doi.org/10.1123/ijspp.2023-0077
69. Kepir E, Demiral F, Akaras E, Paksoy AE, Sevindik Aktas B, Yilmaz Cankaya B, et al. Hamstring muscle stiffness in athletes with and without anterior cruciate ligament reconstruction history: a retrospective study. J Clin Med. 2024;13(15):4370.
70. Clark BC. Neural mechanisms of age-related loss of muscle performance and physical function. J Gerontol Biol Sci Med Sci. 2023;78(Supplement1):8–13. https://doi.org/10.1093/gerona/glad029
71. Aare S, Spendiff S, Vuda M, Elkrief D, Perez A, Wu Q, et al. Failed reinnervation in aging skeletal muscle. Skelet Muscle. 2016;6(1):29. https://doi.org/10.1186/s13395-016-0101-y
72. Carl MA, Magnus T, Alf T. Acute hamstring injuries in Swedish elite football: a prospective randomised controlled clinical trial comparing two rehabilitation

protocols. Br J Sports Med. 2013;47(15):953. https://doi.org/10.1136/bjsports-2013-092165

73. Rey E, Paz-Domínguez Á, Porcel-Almendral D, Paredes-Hernández V, Barcala-Furelos R, Abelairas-Gómez C. Effects of a 10-week nordic hamstring exercise and Russian belt training on posterior lower-limb muscle strength in elite junior soccer players. J Strength Cond Res. 2017;31(5):1198–205. https://doi.org/10.1519/jsc.0000000000001579

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.